

☒ Easy Level (10 Questions)

Q1. What does IDS stand for?

- A) Information Detection System
- B) Intrusion Detection System
- C) Intelligent Defense Software
- D) Internal Data Service

Answer: B

Q2. Which of the following **prevents attacks** in real-time?

- A) IDS
- B) Firewall
- C) IPS
- D) Antivirus

Answer: C

Q3. Which system only monitors and alerts, but does not block traffic?

- A) IPS
- B) IDS
- C) Firewall
- D) Switch

Answer: B

Q4. What is a DoS attack?

- A) Disabling Operating System
- B) Data on System
- C) Denial of Service
- D) Detection of Signatures

Answer: C

Q5. Which of the following uses predefined patterns to detect threats?

- A) Anomaly-based IDS
- B) Signature-based IDS
- C) Behavioral firewall
- D) Manual auditing

Answer: B

Q6. What is a false positive in IDS?

- A) System missed a real threat
- B) System correctly flagged an attack
- C) System flagged normal activity as a threat
- D) System failed to start

Answer: C

Q7. HIDS works by:

- A) Monitoring network traffic
- B) Monitoring host activities like logs and file access
- C) Routing packets
- D) Blocking ports

Answer: B

Q8. NIDS is typically placed:

- A) On a hard disk
- B) At the core of a switch
- C) On the network perimeter
- D) On backup servers

Answer: C

Q9. An attacker floods a server with traffic to bring it down. This is:

- A) Port scan
- B) Brute-force
- C) XSS
- D) DoS attack

Answer: D

Q10. Which detection technique helps in identifying **zero-day attacks**?

- A) Signature-based
- B) Port blocking
- C) Anomaly-based
- D) Static rule sets

Answer: C

☒ Ready for the 20 medium-level questions?

You said:

yes

ChatGPT said:

Great! Here's the next set of **20 Medium-Level MCQs** on IDS / IPS, types of attacks, detection methods, vulnerabilities, and event analysis — with answers and clear explanations.

☒ IDS / IPS – Medium-Level MCQs (20 Questions)

Q11. Which IDS detection method uses **machine learning or statistical models** to establish a baseline of normal behavior?

- A) Signature-based
- B) Anomaly-based
- C) Rule-based
- D) Static detection

Answer: B

☒ *Anomaly-based detection identifies deviations from normal patterns, helping detect unknown threats.*

Q12. Which type of IDS is ideal for detecting unauthorized file modifications on a server?

- A) NIDS
- B) HIDS
- C) IPS
- D) DMZ

Answer: B

☒ *Host-based IDS monitors file integrity, system calls, and logs on individual machines.*

Q13. In which situation would an **IPS** take action but an **IDS** would not?

- A) Scanning log files
- B) Detecting port scans
- C) Blocking a buffer overflow attempt
- D) Sending alert emails

Answer: C

Q14. Which type of attack tries to **discover system vulnerabilities** by scanning open ports?

- A) DoS
- B) Port Scanning
- C) SQL Injection
- D) Rootkit

Answer: B

Q15. What is the purpose of a **Security Event and Incident Management (SEIM)** system in relation to IDS/IPS?

- A) Encrypt VPN tunnels
- B) Store backups
- C) Correlate and analyze alerts from multiple IDS/IPS sources
- D) Block all traffic

Answer: C

Q16. A user receives an alert for a known attack pattern, but no damage is done. This is an example of:

- A) False negative
- B) True positive
- C) False positive
- D) IDS failure

Answer: B

Q17. What technique allows **IPS systems to block** suspicious traffic without human intervention?

- A) Traffic mirroring
- B) Packet rewriting
- C) Inline packet filtering
- D) Tap monitoring

Answer: C

Q18. An attacker performs **SQL Injection** to bypass login. Which tool is more likely to detect it?

- A) NIDS
- B) Switch
- C) HIDS
- D) NAT

Answer: A

☒ *NIDS monitors HTTP payloads to catch SQL keywords being passed in requests.*

Q19. A **hybrid IDS** combines:

- A) IPS and Firewall
- B) Antivirus and VPN
- C) Host-based and network-based IDS
- D) Signature and firewall rules

Answer: C

Q20. Which of the following is the **biggest disadvantage of anomaly-based IDS**?

- A) Can't detect new attacks
- B) Always requires physical hardware
- C) High false positive rate
- D) Needs NAT support

Answer: C

Q21. Why is **signature-based IDS** fast in detection?

- A) It scans encrypted traffic
- B) It only alerts on TCP traffic
- C) It compares traffic to known patterns
- D) It drops all unknown packets

Answer: C

Q22. What kind of vulnerability is caused due to **missing input validation** in code?

- A) Design flaw
- B) Configuration issue
- C) Implementation vulnerability
- D) Protocol misalignment

Answer: C

Q23. If an attacker manipulates packet headers to confuse IDS sensors, this is known as:

- A) Packet flooding
- B) Header injection
- C) Evasion technique
- D) Passive scanning

Answer: C

Q24. Which of the following is a **design flaw**, not an implementation bug?

- A) Allowing root access via UI
- B) Missing brackets in SQL query
- C) Misconfigured firewall
- D) Memory leak in C code

Answer: A

☒ *Design flaws are logic-based — such as poor security policies or trust models.*

Q25. A false negative in an IDS/IPS is when:

- A) It blocks normal traffic
- B) It misclassifies benign activity as malicious
- C) It misses a real attack
- D) It sends too many alerts

Answer: C

Q26. What is a common limitation of encrypted traffic for IDS?

- A) Encrypted traffic is dropped
- B) IDS cannot inspect payloads
- C) Packets are skipped randomly
- D) IP addresses are encrypted

Answer: B

Q27. Snort is an example of a:

- A) HIDS
- B) NIDS
- C) WAF
- D) Firewall

Answer: B

☒ *Snort is an open-source network-based intrusion detection and prevention system.*

Q28. Why might IPS be less suitable in high-speed networks?

- A) Needs two-factor authentication
- B) Slows down encryption
- C) Can introduce latency due to real-time traffic analysis
- D) Doesn't log packets

Answer: C

Q29. Which of the following attacks is most likely detected using behavior-based anomaly detection?

- A) Known trojan
- B) Buffer overflow exploit
- C) Zero-day exploit
- D) Port 80 scan

Answer: C

Q30. What is the role of contextual analysis in IDS alerts?

- A) Blocks traffic automatically
- B) Identifies physical location of source
- C) Determines if an alert is relevant based on system activity
- D) Encrypts alerts

Answer: C

🚧 Up next: **10 Hard-Level Questions** on IDS/IPS. Proceed?

You said:

yes alws proceed

ChatGPT said:

Perfect! Let's now cover the final part:

☒ IDS / IPS – Hard-Level MCQs (10 Questions with Answers & Explanations)

Q31. Which of the following best describes an **evasion technique** used against NIDS?

- A) Sending encrypted traffic
- B) Flooding a switch port
- C) Fragmenting packets to bypass signature matching
- D) Using strong passwords

Answer: C

☒ *Some attackers fragment payloads to bypass NIDS that don't properly reassemble packets.*

Q32. In a high-performance environment, why might you choose a **passive IDS over an inline IPS**?

- A) Passive IDS does not require encryption
- B) Passive IDS does not drop packets, so it ensures zero latency
- C) Passive IDS encrypts all traffic
- D) Passive IDS performs firewall duties

Answer: B

☒ *Inline IPS inspects and may drop packets, introducing latency – which is avoided by passive IDS.*

Q33. Which advanced detection technique involves building profiles for user behavior and flagging deviations?

- A) Signature-based detection
- B) Behavioral analysis
- C) Firewall rule matching
- D) Port mapping

Answer: B

☒ *Behavioral analysis learns baseline activity and flags anomalies (like unexpected logins, access times, etc.).*

Q34. A well-designed IDS/IPS system uses a **correlation engine** to:

- A) Filter duplicate alerts
- B) Encrypt logs
- C) Map MAC addresses
- D) Auto-update virus signatures

Answer: A

☒ *Correlation engines connect multiple alerts to determine the context or pattern of an attack.*

Q35. What is the **main risk** of using only signature-based detection in modern networks?

- A) Incompatibility with VLANs
- B) Misses zero-day or unknown threats
- C) Works only on Linux
- D) Can't detect UDP packets

Answer: B

Q36. Which of the following can help an IPS reduce **false positives** without compromising security?

- A) Switching to only anomaly detection
- B) Ignoring unknown protocols
- C) Context-aware rules with correlation of events
- D) Disabling encryption

Answer: C

☒ *Context-aware logic ensures that alerts are only triggered when meaningful criteria are met (e.g., multiple failed logins from the same source).*

Q37. If an IPS is **not correctly tuned**, it can lead to which operational issue?

- A) Insecure TLS
- B) Log rotation failure
- C) Legitimate traffic being blocked (false positives)
- D) High link utilization

Answer: C

Q38. A **zero-day exploit** targets a vulnerability that is:

- A) Already patched
- B) Known for years
- C) Unknown to vendor and public
- D) Publicly disclosed but unused

Answer: C

☒ *Zero-day attacks exploit previously unknown vulnerabilities.*

Q39. What makes **encrypted malicious payloads** difficult to detect for IDS/IPS?

- A) They use unknown TCP ports
- B) They contain random bytes
- C) IDS/IPS cannot inspect inside encrypted content without decryption
- D) They use HTTP/2

Answer: C

Q40. Which of the following describes an **implementation vulnerability**?

- A) Allowing default admin credentials to remain
- B) Flawed trust model in system architecture
- C) Using HTTP instead of HTTPS by design
- D) Absence of access control in source code

Answer: D

☒ *Implementation flaws are typically code-level bugs, such as missing validation, improper permissions, etc.*